Azure Infrastructure Review

Tenant contoso.onmicrosoft.com, three subscriptions

Tenant	Subscriptions	Period
contoso.onmicrosoft.com	**contoso-prod, contoso-nonprod, contoso-shared**	**October 2026**

Generated

2026-11-02 07:27

Generated

2026-11-02 07:27:09 +01:00

! Critical ▾ Show the rules

! RDP (3389) and SQL (1433) are open to the internet on two production network security groups.

Overview

Summary 6

OVERALL

Needs attention

Two internet-facing rules and rising cost need action.

2 Critical | 5 Warning | 3 Info



SECURE SCORE

Fair

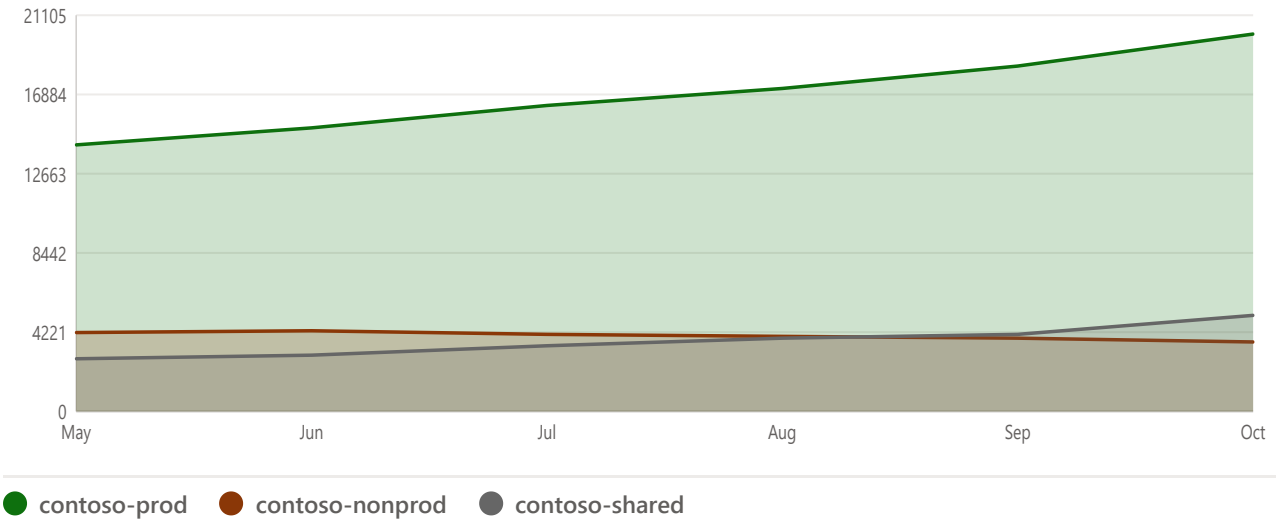
212 recommendations · 37 unhealthy

▲ +3 (+4.4%)

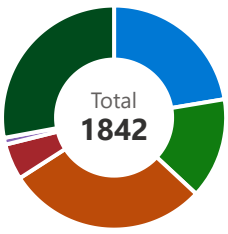
1,842	€28,920	214	96
▲ +52 (+2.9%)	▲ +2520 (+9.5%)	▼ -26 (-10.8%)	UNTAGGED
RESOURCES	COST THIS MONTH	NON-COMPLIANT RESOURCES	



Cost per month (EUR)



Resources by type



Virtual machines	**412** (22%)
Storage accounts	**268** (15%)
Network	**534** (29%)
Databases	**96** (5%)
Kubernetes	**18** (1%)
Other	**514** (28%)

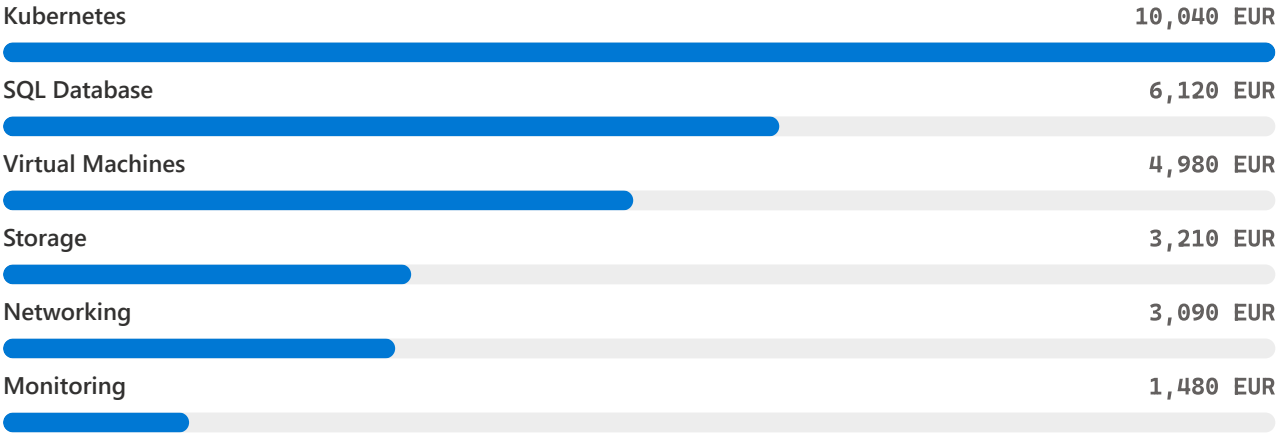
Cost

Summary 4

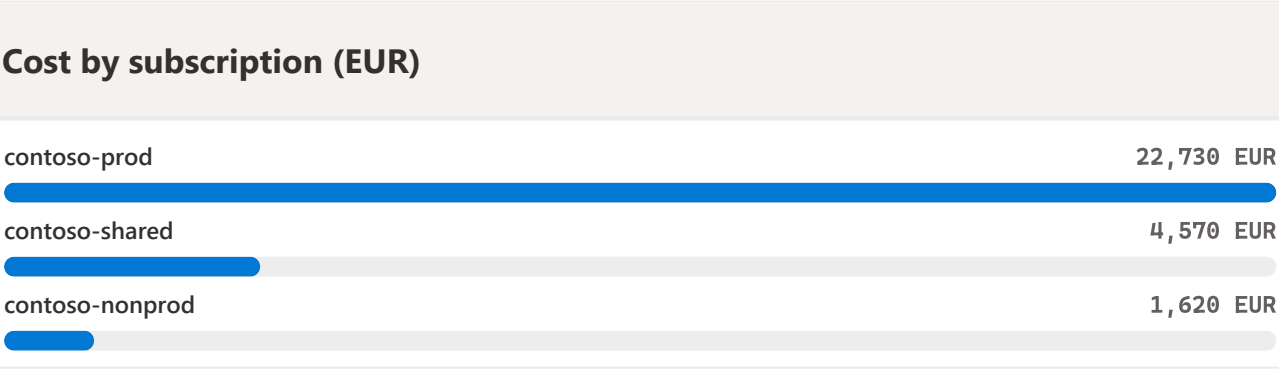
€28,920	€41,200	82%	9.5
MONTH TO DATE	FORECAST	BUDGET USED	CHANGE VS LAST MONTH

Cost › Trend

Cost by service (EUR)



Cost by subscription (EUR)



Cost › Top resources

Most expensive resources

RESOURCE	SUBSCRIPTION	SERVICE	COST (EUR)	6 MONTHS
aks-prod-weu	contoso-prod	Kubernetes	€8,420	
sql-erp-prod	contoso-prod	SQL Database	€6,120	
vmss-web-prod	contoso-prod	Virtual Machines	€4,980	
stdatalakeprod	contoso-prod	Storage	€3,210	
appgw-hub	contoso-shared	Networking	€1,840	
aks-dev-weu	contoso-nonprod	Kubernetes	€1,620	
log-analytics-shared	contoso-shared	Monitoring	€1,480	
fw-hub-premium	contoso-shared	Networking	€1,250	
			€28,920	

Compute

Summary 4

36	26	10	3
VIRTUAL MACHINES	RUNNING	DEALLOCATED OR STOPPED	KUBERNETES CLUSTERS

Compute › Virtual machines

Virtual machines

VM	SUBSCRIPTION	REGION	SIZE	OPERATING SYSTEM	POWER	CPU AVG %	COST / MONTH	TAGS
vm-nonprod-01	contoso-nonprod	northeurope	Standard_D8s_v5	Ubuntu 22.04	RUNNING	11	€472	env:nonprod owner:ok
vm-shared-02	contoso-shared	italynorth	Standard_E8s_v5	Ubuntu 22.04	RUNNING	21	€812	env:nonprod owner:ok
vm-prod-03	contoso-prod	westeurope	Standard_B2ms	Ubuntu 22.04	RUNNING	34	€416	env:prod owner:ok
vm-nonprod-04	contoso-nonprod	northeurope	Standard_F8s_v2	Windows Server 2022	RUNNING	23	€674	env:nonprod owner:ok
vm-shared-05	contoso-shared	italynorth	Standard_D4s_v5	Ubuntu 22.04	RUNNING	73	€938	env:nonprod owner:ok
vm-prod-06	contoso-prod	westeurope	Standard_D8s_v5	Ubuntu 22.04	RUNNING	51	€826	env:prod owner:ok

VM	SUBSCRIPTION	REGION	SIZE	OPERATING SYSTEM	POWER	CPU AVG %	COST / MONTH	TAGS
vm-nonprod-07	contoso-nonprod	northeurope	Standard_E8s_v5	Ubuntu 22.04	DEALLOCATED	56	€413	env:nonprod owner:missing
vm-shared-08	contoso-shared	italynorth	Standard_B2ms	Windows Server 2022	RUNNING	83	€283	env:nonprod owner:ok
vm-prod-09	contoso-prod	westeurope	Standard_F8s_v2	Ubuntu 22.04	DEALLOCATED	29	€495	env:prod owner:ok
vm-nonprod-10	contoso-nonprod	northeurope	Standard_D4s_v5	Ubuntu 22.04	RUNNING	78	€264	env:nonprod owner:ok
vm-shared-11	contoso-shared	italynorth	Standard_D8s_v5	Ubuntu 22.04	STOPPED	69	€432	env:nonprod owner:ok
vm-prod-12	contoso-prod	westeurope	Standard_E8s_v5	Windows Server 2022	DEALLOCATED	46	€132	env:prod owner:ok
vm-nonprod-13	contoso-nonprod	northeurope	Standard_B2ms	Ubuntu 22.04	RUNNING	62	€680	env:nonprod owner:ok
vm-shared-14	contoso-shared	italynorth	Standard_F8s_v2	Ubuntu 22.04	RUNNING	66	€418	env:nonprod owner:missing

VM	SUBSCRIPTION	REGION	SIZE	OPERATING SYSTEM	POWER	CPU AVG %	COST / MONTH	TAGS
vm-prod-15	contoso-prod	westeurope	Standard_D4s_v5	Ubuntu 22.04	RUNNING	54	€176	env:prod owner:ok
vm-nonprod-16	contoso-nonprod	northeurope	Standard_D8s_v5	Windows Server 2022	DEALLOCATED	34	€397	env:nonprod owner:ok
vm-shared-17	contoso-shared	italynorth	Standard_E8s_v5	Ubuntu 22.04	RUNNING	11	€362	env:nonprod owner:ok
vm-prod-18	contoso-prod	westeurope	Standard_B2ms	Ubuntu 22.04	DEALLOCATED	43	€743	env:prod owner:ok
vm-nonprod-19	contoso-nonprod	northeurope	Standard_F8s_v2	Ubuntu 22.04	RUNNING	69	€622	env:nonprod owner:ok
vm-shared-20	contoso-shared	italynorth	Standard_D4s_v5	Windows Server 2022	RUNNING	34	€211	env:nonprod owner:ok
vm-prod-21	contoso-prod	westeurope	Standard_D8s_v5	Ubuntu 22.04	RUNNING	68	€314	env:prod owner:missing
vm-nonprod-22	contoso-nonprod	northeurope	Standard_E8s_v5	Ubuntu 22.04	RUNNING	53	€723	env:nonprod owner:ok

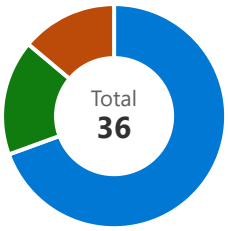
VM	SUBSCRIPTION	REGION	SIZE	OPERATING SYSTEM	POWER	CPU AVG %	COST / MONTH	TAGS
vm-shared-23	contoso-shared	italynorth	Standard_B2ms	Ubuntu 22.04	STOPPED	52	€102	env:nonprod owner:ok
vm-prod-24	contoso-prod	westeurope	Standard_F8s_v2	Windows Server 2022	DEALLOCATED	60	€441	env:prod owner:ok
vm-nonprod-25	contoso-nonprod	northeurope	Standard_D4s_v5	Ubuntu 22.04	RUNNING	9	€453	env:nonprod owner:ok
vm-shared-26	contoso-shared	italynorth	Standard_D8s_v5	Ubuntu 22.04	RUNNING	19	€889	env:nonprod owner:ok
vm-prod-27	contoso-prod	westeurope	Standard_E8s_v5	Ubuntu 22.04	STOPPED	14	€140	env:prod owner:ok
vm-nonprod-28	contoso-nonprod	northeurope	Standard_B2ms	Windows Server 2022	RUNNING	79	€773	env:nonprod owner:missing
vm-shared-29	contoso-shared	italynorth	Standard_F8s_v2	Ubuntu 22.04	RUNNING	74	€104	env:nonprod owner:ok
vm-prod-30	contoso-prod	westeurope	Standard_D4s_v5	Ubuntu 22.04	RUNNING	45	€192	env:prod owner:ok

VM	SUBSCRIPTION	REGION	SIZE	OPERATING SYSTEM	POWER	CPU AVG %	COST / MONTH	TAGS
vm-nonprod-31	contoso-nonprod	northeurope	Standard_D8s_v5	Ubuntu 22.04	RUNNING	35	€117	env:nonprod owner:ok
vm-shared-32	contoso-shared	italynorth	Standard_E8s_v5	Windows Server 2022	RUNNING	23	€834	env:nonprod owner:ok
vm-prod-33	contoso-prod	westeurope	Standard_B2ms	Ubuntu 22.04	RUNNING	8	€158	env:prod owner:ok
vm-nonprod-34	contoso-nonprod	northeurope	Standard_F8s_v2	Ubuntu 22.04	STOPPED	48	€194	env:nonprod owner:ok
vm-shared-35	contoso-shared	italynorth	Standard_D4s_v5	Ubuntu 22.04	STOPPED	48	€953	env:nonprod owner:missing
vm-prod-36	contoso-prod	westeurope	Standard_D8s_v5	Windows Server 2022	RUNNING	79	€859	env:prod owner:ok

VMs by size



Power state



Running	**25** (69%)
Deallocated	**6** (17%)
Stopped	**5** (14%)

Compute › Kubernetes

AKS clusters

aks-prod-weu
contoso-prod

1.29.4	18
VERSION	NODES

Cluster

Kubernetes version
1.29.4

Node pools
system, apps, gpu

Nodes
18

Checks

✓ Supported version	Up to date
✓ Autoscaler on	
✓ Private API server	

aks-dev-weu

contoso-nonprod

1.27.9 VERSION | 6 NODES

Cluster

Kubernetes version	Node pools
1.27.9	system, apps

Nodes
6

Checks

- ✗ Supported version — Kubernetes 1.27 leaves support in 3 weeks
- ✓ Autoscaler on
- ✗ Private API server

aks-data-neu

contoso-shared

1.29.4 VERSION | 9 NODES

Cluster

Kubernetes version	Node pools
1.29.4	system, spark

Nodes
9

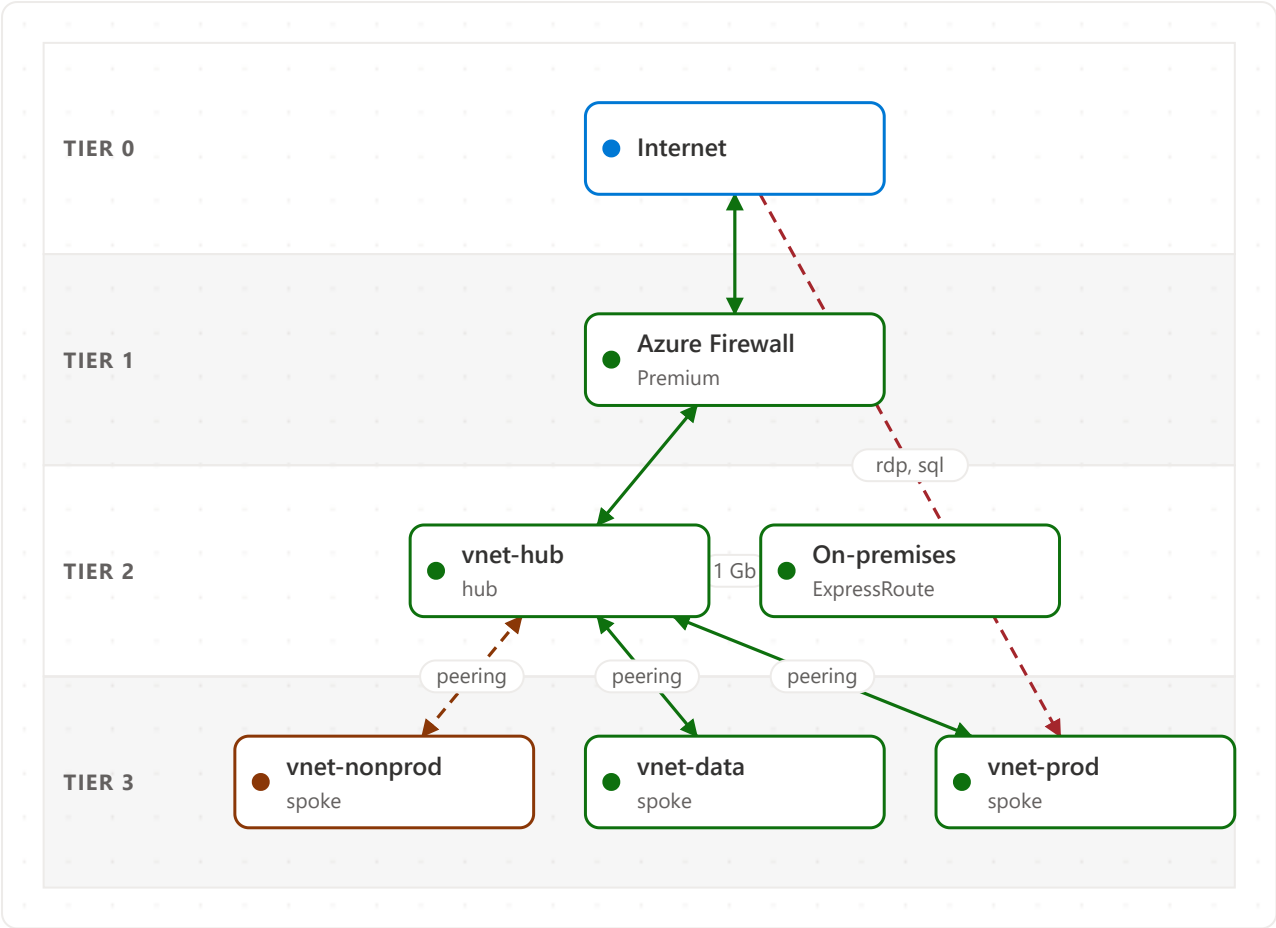
Checks

- ✓ Supported version — Up to date
- ✓ Autoscaler on
- ✓ Private API server

Network › Topology

Hub and spoke topology

OK · Warning · Critical · Info · Unknown · link · dashed

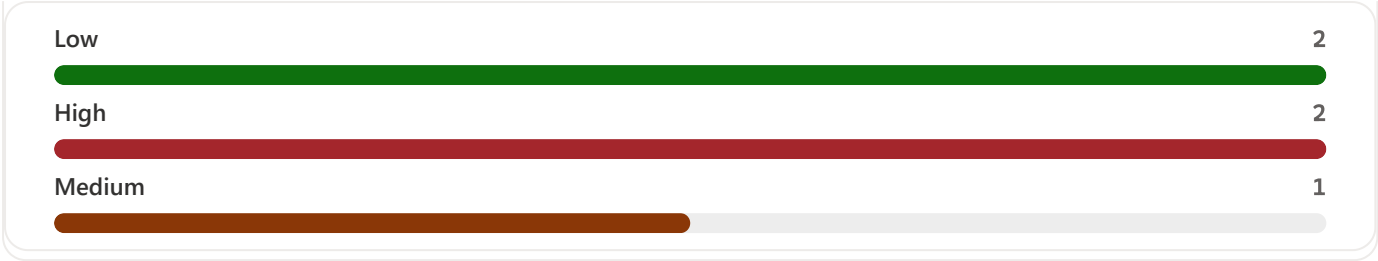


Network › Exposure

Network security group rules reachable from the internet

NSG	SUBSCRIPTION	RULE	PORT	SOURCE	RISK
nsg-web-prod	contoso-prod	allow-https-any	443	Internet	LOW
nsg-jump-prod	contoso-prod	allow-rdp-any	3389	Internet	HIGH
nsg-db-prod	contoso-prod	allow-sql-any	1433	Internet	HIGH
nsg-dev	contoso-nonprod	allow-ssh-any	22	Internet	MEDIUM
nsg-shared	contoso-shared	allow-dns	53	VirtualNetwork	LOW

Rules by risk



Storage

Storage accounts

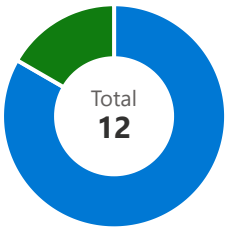
ACCOUNT	SUBSCRIPTION	REDUNDANCY	PUBLIC ACCESS	MINIMUM TLS	SIZE (TB)
stdev01	contoso-nonprod	ZRS	DISABLED	1.2	14
stsh02	contoso-shared	GRS	DISABLED	1.2	3.5
stprod03	contoso-prod	GZRS	DISABLED	1.2	2.9
stdev04	contoso-nonprod	LRS	DISABLED	1.2	16.7
stsh05	contoso-shared	ZRS	ENABLED	1.2	11.1
stprod06	contoso-prod	GRS	DISABLED	1.0	7.5
stdev07	contoso-nonprod	GZRS	DISABLED	1.2	10.4
stsh08	contoso-shared	LRS	DISABLED	1.2	17
stprod09	contoso-prod	ZRS	DISABLED	1.2	15.8
stdev10	contoso-nonprod	GRS	ENABLED	1.2	16.9
stsh11	contoso-shared	GZRS	DISABLED	1.2	6.6
stprod12	contoso-prod	LRS	DISABLED	1.0	14.7
					137.10

Accounts by redundancy





Public access



Disabled	10 (83%)
Enabled	2 (17%)

Security

Summary 4

37 ▼ -7 (-15.9%) UNHEALTHY RECOMMENDATIONS	6 HIGH SEVERITY	2 OPEN TO THE INTERNET	7 of 9 DEFENDER PLANS ON

Security › Recommendations

Defender for Cloud recommendations

CRITICAL New **Management ports are open to the internet** Network

Description

RDP and SQL are reachable from any address on two production network security groups.

Recommendation

Remove the rules and use Azure Bastion and private endpoints.

Affected NSGs	Exposure
2	Internet

Evidence (2)

OBJECT	VALUE	STATUS
nsg-jump-prod	3389 from Internet	Critical
nsg-db-prod	1433 from Internet	Critical

WARNING A cluster runs a Kubernetes version leaving support soon

Kubernetes

Recommendation

Upgrade to 1.29 before support ends.

Affected (1)

`aks-dev-weu`

WARNING Storage accounts allow public access or old TLS

Storage

Recommendation

Disable public blob access and require TLS 1.2.

Evidence (4)

OBJECT	VALUE	STATUS
stsh05	public: Enabled, TLS 1.2	Warning
stprod06	public: Disabled, TLS 1.0	Warning
stdev10	public: Enabled, TLS 1.2	Warning
stprod12	public: Disabled, TLS 1.0	Warning

WARNING Worse Virtual machines are missing the owner tag

Compute

Recommendation

Tag every VM with an owner so alerts can reach someone.

Evidence (5)

OBJECT	VALUE	STATUS
vm-nonprod-07	owner tag missing	Warning
vm-shared-14	owner tag missing	Warning
vm-prod-21	owner tag missing	Warning
vm-nonprod-28	owner tag missing	Warning
vm-shared-35	owner tag missing	Warning

INFO Resolved **Two service principals have secrets older than a year** Identity

No further detail.

Security › Posture

Summary 2



SECURE SCORE

Fair

▲ +3 (+4.4%)

CLOUD SECURITY MATURITY

Defined

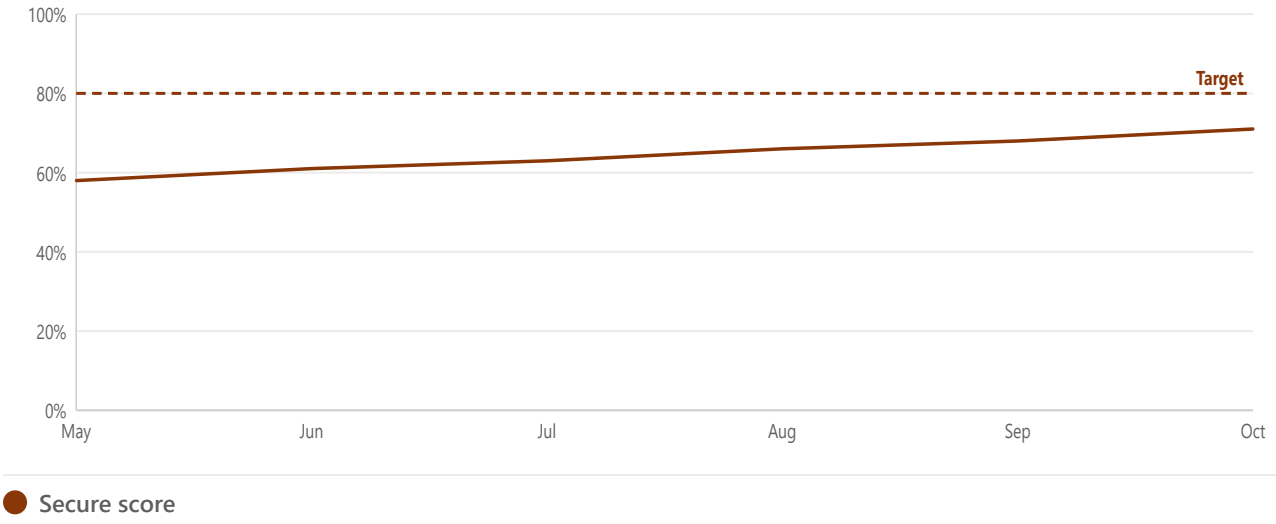
Level 3 of 5

1 Initial · 2 Managed · 3 **Defined** · 4 Measured · 5 Optimised

To reach level 4 (Measured)

- ○ No internet-facing management ports
- ○ All subscriptions on the secure score target
- ✓ Measure policy compliance every week

Secure score, last six months



Governance › Policy

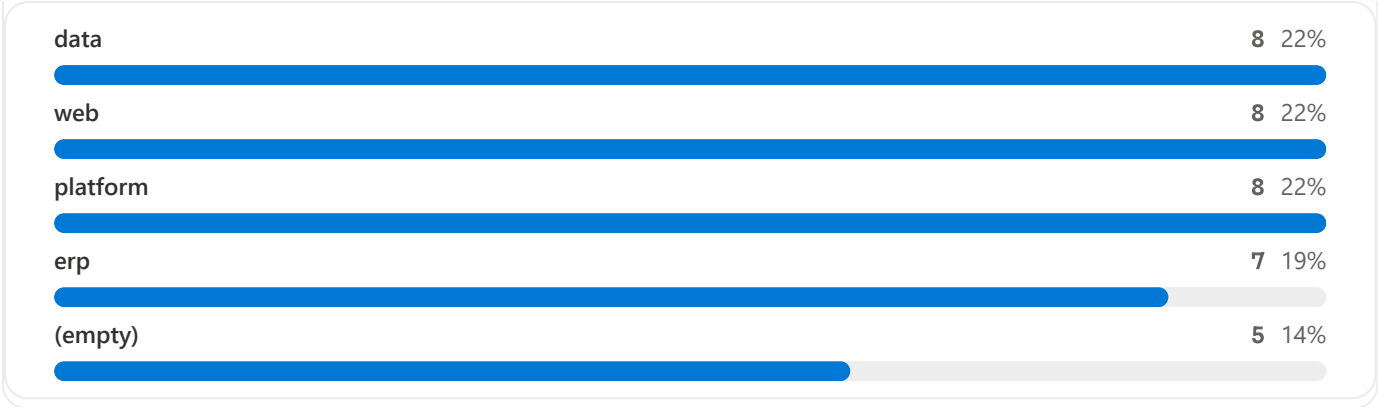
Policy compliance (%)

Subscription Initiative	Azure Security Benc...	CIS 2.0	ISO 27001	Tagging standard
contoso-prod	88	79	84	62
contoso-nonprod	74	66	70	41
contoso-shared	93	90	91	78

41.0% 93.0%

Governance › Tags

VMs by owner tag



Resources without an owner

VM	SUBSCRIPTION	REGION	SIZE	COST / MONTH
vm-nonprod-07	contoso-nonprod	northeurope	Standard_E8s_v5	€413
vm-shared-14	contoso-shared	italynorth	Standard_F8s_v2	€418
vm-prod-21	contoso-prod	westeurope	Standard_D8s_v5	€314
vm-nonprod-28	contoso-nonprod	northeurope	Standard_B2ms	€773
vm-shared-35	contoso-shared	italynorth	Standard_D4s_v5	€953